Cross-field Consequences of Bessel-Value Independence: Circular Mixtures, Random-Time Quantum Walks, and Robin Spectra

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Abstract

We derive applications in statistics, spectral graph theory, and elliptic spectral theory from explicitly stated arithmetic inputs of the unified Euler–Bessel manuscript. For finite von Mises mixtures with algebraic weights, positive algebraic concentrations, and algebraic unit directions, the first r circular moments identify the entire mixing measure when each concentration supports at most r directions. This bound is sharp and independent of the number of concentrations; its inverse is nevertheless discontinuous. We also exclude all-algebraic interior likelihood critical points. For finite algebraic Hermitian Hamiltonians sampled at beta-distributed random times of algebraic half-width, every averaged mixing eigenvalue other than 1 is transcendental over the field of algebraic exponentials; all principal submatrices are nonsingular. For Robin problems on balls, integral-intersection and unit inputs exclude positive algebraic eigenvalues for a wider ring of boundary parameters, with explicit negative exceptions in odd dimensions. A recurrence argument separately removes accidental angular degeneracy for every algebraic Robin parameter, allowing the Hassannezhad–Sher Pleijel theorem to apply without a genericity restriction in that arithmetic subclass. We distinguish these applications from the general open problems and make no claim of established historical priority.

Status and dependencies. The arguments below are complete deductions from the stated inputs of [1], a working manuscript rather than a published theorem source. The statistical and quantum-walk results use only Input 1.1. The stronger Robin results also use the ring statements in (8). The algebraic-parameter nondegeneracy proof is a classical recurrence/transcendence argument. Literature comparison is limited to the cited primary sources and does not establish novelty. In particular no claim is made to solve the unrestricted random-time quantum-walk programme, the general-domain Pleijel question, or the clamped-plate multiplicity conjecture.

1 Arithmetic input and statistical conventions

Write $k = \overline{\mathbb{Q}}$ and $K_{\exp} = \overline{\mathbb{Q}}(e^a : a \in \overline{\mathbb{Q}})$. We isolate precisely the value-independence statement used here.

Input 1.1 (Joint Bessel values from the unified manuscript)**.** For any nonzero algebraic numbers $t_1, \dots, t_m$ distinct modulo sign, the $2m$ numbers $I_0(t_j), I_1(t_j)$ are jointly algebraically independent over $K_{\exp}$.

This is a subfamily of [1, Theorem 3.3]. The cited proof uses the product SL_2^m differential Galois group of the distinct Bessel scales, its separation from solvable exponential systems, and Siegel–Shidlovskii specialization at an ordinary algebraic point. The present note proves applications of this input; it does not reproduce or independently certify that foundational proof. For positive distinct algebraic t_j, the birational change of coordinates gives the equivalent independence of

$$I_0(t_1), C(t_1), \dots, I_0(t_m), C(t_m), \qquad C(t) = I_1(t)/I_0(t). \tag{1}$$

Consequently $1, C(t_1), \dots, C(t_m)$ are linearly independent over $\overline{\mathbb{Q}}$, and each $C(t_j)$ is transcendental over $K_{\exp}(I_0(t_1), \dots, I_0(t_m))$. The statistical results use this positive-scale consequence. The quantum-walk application also uses imaginary scales. Section 7 states the two additional ring-theoretic inputs needed for its stronger Robin spectral assertions.

Identify $\mathbb{S}^1$ either with complex numbers of modulus one or with real unit vectors in $\mathbb{R}^2$. For $u \in \mathbb{S}^1$ and $t > 0$, let $\mathrm{VM}(u, t)$ have density, relative to uniform probability measure on the circle,

$$f_{u,t}(z) = \frac{\exp(t \operatorname{Re}(\overline{u}z))}{I_0(t)}.$$

Its complex circular moments are

$$\int_{\mathbb{S}^1} z^n \, d\,\mathrm{VM}(u,t)(z) = u^n R_n(t), \qquad R_n(t) = I_n(t)/I_0(t), \quad n \geq 0. \tag{2}$$

This follows directly by rotating the standard integral for I_n. The concentration MLE and its Bessel-ratio score are standard; see Hornik–Grün [10]. Ordinary finite-mixture identifiability is already classical [8, 9]. Our question is how many exact moments suffice under explicit arithmetic restrictions.

An *arithmetic mixture* means

$$P = \sum_{j=1}^{m} w_j \,\mathrm{VM}(u_j, t_j), \qquad w_j \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}, \quad \sum_j w_j = 1, \quad u_j \in \overline{\mathbb{Q}} \cap \mathbb{S}^1, \quad t_j \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}. \tag{3}$$

Repeated identical pairs (u_j, t_j) are merged. The direction condition means that the Cartesian coordinates of u_j are algebraic; it does *not* mean that an angle chosen to represent u_j is algebraic. Equality of mixtures below means equality of their discrete mixing measures, up to permutation, with identical pairs merged.

2 A sharp moment bound independent of the number of concentrations

Lemma 2.1 (Separation of concentration groups)**.** *Let two arithmetic mixtures have equal nth complex circular moment, where $n \geq 1$. For each positive concentration t occurring in either mixture, their grouped directional moments*

$$a_{t,n} := \sum_{j : t_j = t} w_j u_j^n$$

are equal. An absent group has all its moments zero.

Proof. The Bessel recurrence gives

$$R_{n+1}(t) = R_{n-1}(t) - \frac{2n}{t} R_n(t), \qquad R_0 = 1, \quad R_1 = C(t).$$

Thus $R_n(t) = A_n(t) + B_n(t)C(t)$, with $A_n, B_n \in \mathbb{Q}[t^{-1}]$, $B_0 = 0$, and $B_1 = 1$. For $t > 0$, $B_n(t)$ has sign $(-1)^{n-1}$ for every $n \geq 1$. Indeed $B_2 = -2/t < 0$, and in the recurrence for B_{n+1} the two terms B_{n-1} and $-(2n/t)B_n$ have the same sign. In particular, $B_n(t) \neq 0$.

Write $\Delta a_{t,n}$ for the difference of the two grouped moments. The assumed equality is

$$0 = \sum_t A_n(t)\Delta a_{t,n} + \sum_t B_n(t)\Delta a_{t,n}C(t).$$

All coefficients are algebraic. Linear independence of 1 and the distinct $C(t)$, from (1), gives $B_n(t)\Delta a_{t,n} = 0$ for each t, proving the assertion. □

Lemma 2.2 (Positive circular atomic measures with unknown mass)**.** *Two finite positive measures on $\mathbb{S}^1$, each supported on at most r points, are equal if their moments of orders $1, \ldots, r$ are equal. Their total masses need not be specified in advance.*

Proof. Write $a_n = \int u^n \, d\nu(u)$ and $a_{-n} = \overline{a_n}$. The Toeplitz matrix

$$T = (a_{j-i})_{0 \leq i,j \leq r} = a_0 I_{r+1} + H$$

is positive semidefinite and has rank at most r, because it is a sum of at most r positive rank-one matrices. The Hermitian matrix H is determined by $a_1, \ldots, a_r$ and has zero diagonal. Singularity and positivity therefore force

$$a_0 = -\lambda_{\min}(H).$$

The two measures have the same total mass as well as moments $-r, \ldots, r$. Their signed difference is supported at $s \leq 2r$ distinct nonzero nodes $v_1, \ldots, v_s$. Using any s consecutive exponents in $[-r, r]$ gives a Vandermonde system with nonzero determinant. Every signed weight is zero. The argument also includes the zero measure. This is a direct form of the classical positive Toeplitz/Vandermonde mechanism; compare [11]. □

Theorem 2.3 (Sharp arithmetic finite-moment identifiability)**.** *Fix an integer $r \geq 1$. Among arithmetic mixtures having at most r distinct directions at each concentration, the first r complex circular moments determine the entire mixing measure. The total number of distinct concentrations may be any finite number and is not assumed known. The bound r cannot uniformly be replaced by $r - 1$.*

Proof. Suppose two mixtures have the same moments of orders $1, \ldots, r$. Lemma 2.1 gives equality of these grouped directional moments at each concentration in the union of their concentration sets. At each fixed concentration, the positive directional measures have at most r atoms. Lemma 2.2 makes them equal, including their masses. This also rules out a concentration appearing in just one mixture. All components and weights therefore agree.

For sharpness take concentration $t = 1$, equal weights $1/r$ on the rth roots of unity, and compare with the same configuration rotated by $\zeta_{2r} = e^{\pi i/r}$. For both mixtures the moments of orders $1, \ldots, r-1$ vanish, whereas their rth moments are respectively $R_r(1)$ and $-R_r(1)$. These differ because $I_r(1) > 0$. For $r = 1$, the assertion that zero moments cannot identify a distribution is interpreted literally. □

Corollary 2.4 (One complex moment)**.** *If an arithmetic mixture has no repeated concentration, its first complex circular moment determines its number of components, all its concentrations, all its directions, and all its weights within this class. In particular this is an exact injectivity theorem for mixtures with arbitrarily many components.*

Proof. Apply Theorem 2.3 with $r = 1$. More directly, linear independence of the $C(t)$ determines the coefficient wu attached to each concentration. Since $w > 0$ and $|u| = 1$, it determines $w = |wu|$ and then u. □

Remark 2.5 (What is new in the transfer). Finite von Mises mixture identifiability from the entire density is not a new result. The arithmetic separation in Lemma 2.1, combined with the classical positive moment argument, gives the sharper conclusion from only r moments under the stated restrictions. The arithmetic input is essential to that conclusion. A literature search did not locate this exact formulation; this is not a claim that a previously published open conjecture has been resolved.

3 Why exact identification does not give stable recovery

Proposition 3.1 (Discontinuity already for one moment)**.** *On the class in Corollary 2.4, the inverse of the first-moment map need not be continuous, even if the output is measured as a probability density in total variation.*

Proof. Choose positive algebraic $a < c < b$ and the common direction $u = 1$. The function C is strictly increasing, since $C'(t) = \mathrm{Var}_{\mathrm{VM}(1,t)}(\mathrm{Re}\, z) > 0$. Put

$$w_* = \frac{C(b) - C(c)}{C(b) - C(a)} \in (0, 1),$$

and choose rational $w_n \in (0, 1)$ tending to w_*. The arithmetic mixtures $P_n = w_n \,\mathrm{VM}(1, a) + (1 - w_n)\,\mathrm{VM}(1, b)$ have first moments tending to the first moment of $P = \mathrm{VM}(1, c)$. Their densities converge uniformly to that of $P_* = w_* \,\mathrm{VM}(1, a) + (1 - w_*)\,\mathrm{VM}(1, b)$. This limiting density differs from that of P: otherwise a nontrivial linear relation among e^{ax}, e^{cx}, e^{bx} would hold for $-1 \le x \le 1$, contrary to linear independence of distinct exponentials. Hence $\|P_n - P\|_{\mathrm{TV}} \to \|P_* - P\|_{\mathrm{TV}} > 0$. □

Thus the theorem gives no stable algorithm for reconstruction from rounded moments, no sample-complexity bound, and no effective separation bound. Such quantitative statements require further arithmetic input, for example explicit lower bounds with degree and height restrictions.

4 Transcendental coordinates at likelihood critical points

For observations $x_1, \ldots, x_N \in \mathbb{S}^1 \subset \mathbb{R}^2$, use the relative-uniform density and write

$$\ell(w, u, t) = \sum_{l=1}^{N} \log \left(\sum_{j=1}^{m} w_j \frac{e^{t_j u_j \cdot x_l}}{I_0(t_j)} \right).$$

An interior point means $w_j > 0$, $\sum_j w_j = 1$, $t_j > 0$, with $u_j \in \mathbb{S}^1$. Concentrations are independently variable, and repeated concentration values at a point are allowed.

Theorem 4.1 (No all-algebraic interior critical point)**.** *Suppose $N \ge 1$ and every Cartesian coordinate of every observation x_l is algebraic. Then every interior critical point of ℓ has at least one transcendental coordinate among its weights, concentrations, and Cartesian direction coordinates.*

Proof. Assume all those parameter coordinates are algebraic at an interior critical point. Let T be the finite set of its distinct concentrations, and put

$$F = K_{\exp}(I_0(t) : t \in T).$$

Each exponent $t_j u_j \cdot x_l$ is algebraic. Therefore every responsibility

$$\rho_{lj} = \frac{w_j e^{t_j u_j \cdot x_l}/I_0(t_j)}{\sum_h w_h e^{t_h u_h \cdot x_l}/I_0(t_h)}$$

belongs to F and is strictly positive. Differentiating in t_j gives

$$\frac{\partial \ell}{\partial t_j} = \sum_l \rho_{lj}(u_j \cdot x_l) - C(t_j) \sum_l \rho_{lj}.$$

The first sum belongs to F and the second coefficient belongs to $F \cap \mathbb{R}_{>0}$. A zero derivative would force $C(t_j) \in F$, contrary to (1). In fact none of the concentration derivatives can vanish at an all-algebraic interior point. □

Corollary 4.2 (Penalized likelihood)**.** *Let p be a differentiable penalty such that all its concentration derivatives are algebraic whenever the full parameter tuple is algebraic. Then Theorem 4.1 also holds for $\ell + p$. In particular it holds for*

$$p = -\psi \sum_j t_j, \qquad \psi \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}.$$

Proof. The additional derivative belongs to $\overline{\mathbb{Q}} \subset F$ and changes only the first sum in the proof. The positive coefficient of $C(t_j)$ is unchanged. □

The linear penalty is the form used by Ng [13, Section 4.3], including $\psi = \zeta/N$ with positive algebraic ζ. Only interior critical points are covered: boundary optima, zero weights, and $t_j = 0$ require separate treatment. Unpenalized mixture likelihoods can be unbounded [13, Theorem 3.1], so existence of a global ordinary MLE is not asserted. The conclusion is that at least one coordinate is transcendental, not that every concentration or every coordinate is transcendental. It is a circular-distribution analogue of the arithmetic perspective of Améndola–Drton–Sturmfels on Gaussian mixture likelihoods [12].

5 Boundaries and further questions

The moment theorem and the likelihood obstruction use different parts of (1): linear independence of the ratios suffices for moment separation, whereas the likelihood argument needs independence of the ratios from all raw normalizing constants and algebraic exponentials. They do not prove a stable mixture estimator, classify likelihood critical points, establish D-finiteness or non-D-finiteness of the inverse Bessel-ratio function, or settle general real-parameter finite-moment identifiability.

A concrete next question is quantitative: with bounds on degrees, heights, component weights and concentration separation, can one give effective lower bounds for the difference of the first r moment vectors of two distinct arithmetic mixtures? The qualitative independence input alone supplies no such bound, and Proposition 3.1 precludes a uniform result without additional restrictions.

6 Arithmetic spectra of random-time quantum walks

Let $H \in M_N(\overline{\mathbb{Q}})$ be Hermitian, and put

$$U_H(t) = e^{itH}, \qquad M_H(t) = U_H(t) \circ \overline{U_H(t)},$$

where $\circ$ denotes entrywise multiplication. For an integer $\nu \geq 0$ and $a > 0$, consider the probability density

$$f_{a,\nu}(r) = \frac{\Gamma(\nu+1)}{a\sqrt{\pi}\,\Gamma(\nu+\frac{1}{2})}\left(1 - \frac{r^2}{a^2}\right)^{\nu-1/2} \mathbf{1}_{(-a,a)}(r). \tag{4}$$

The cases $\nu = 0, 1$ are the arcsine and semicircle laws, respectively. Write $\widehat{M}_{a,\nu}(H) = \mathbb{E}[M_H(R)]$ for this sampling law. Throughout this section $K_{\exp} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}})$.

Theorem 6.1 (Arithmetic spectrum under beta sampling)**.** *Let $a > 0$ be algebraic and $\nu \geq 0$ an integer. Every eigenvalue of $\widehat{M}_{a,\nu}(H)$ algebraic over $K_{\exp}$ equals 1. In particular, $\widehat{M}_{a,\nu}(H)$ is nonsingular. If the support graph of H, with edges uv whenever $u \neq v$ and $H_{uv} \neq 0$, has c connected components, eigenvalue 1 has multiplicity exactly c. Every other eigenvalue is real, belongs to $(-1, 1)$, and is transcendental over $K_{\exp}$.*

Proof. The Poisson integral for Bessel functions [4, §10.9] gives the characteristic function

$$\phi_{a,\nu}(s) = \mathbb{E}[e^{isR}] = \frac{2^\nu \nu!}{(as)^\nu} J_\nu(as) \quad (s \neq 0), \qquad \phi_{a,\nu}(0) = 1. \tag{5}$$

For distinct positive algebraic x_j, Theorem 1.1, applied at ix_j, makes all pairs $(J_0(x_j), J_1(x_j))$ independent over $K_{\exp}$. The recurrence $J_{n+1}(x) = (2n/x)J_n(x) - J_{n-1}(x)$ expresses $J_\nu(x)$ as a nonzero algebraic linear form in its pair: successive-pair recurrence matrices have determinant 1. Therefore, for distinct positive algebraic Δ, the values $z_\Delta = \phi_{a,\nu}(\Delta)$ are jointly algebraically independent over $K_{\exp}$.

Write $H = \sum_r \theta_r E_r$ using distinct eigenvalues. All θ_r and projector entries are algebraic, since $E_r = \prod_{s \neq r}(H - \theta_s I)/(\theta_r - \theta_s)$. Let D be the set of distinct positive gaps $|\theta_r - \theta_s|$, and define real symmetric algebraic matrices

$$B_0 = \sum_r E_r \circ \overline{E_r},$$
$$B_\Delta = \sum_{\substack{r<s \\ |\theta_r - \theta_s| = \Delta}} (E_r \circ \overline{E_s} + E_s \circ \overline{E_r}).$$

The characteristic-function expansion of the average, as in [3, Proposition 13], gives

$$\widehat{M}_{a,\nu}(H) = P(z), \qquad P(Z) = B_0 + \sum_{\Delta \in D} Z_\Delta B_\Delta, \qquad P(\mathbf{1}) = I_N. \tag{6}$$

The last identity follows from $\sum_r E_r = I_N$.

If an eigenvalue ρ is algebraic over $K_{\exp}$, the tuple z remains algebraically independent over $L = K_{\exp}(\rho)$. Indeed $L/K_{\exp}$ and $L(z)/K_{\exp}(z)$ are algebraic, so $\operatorname{trdeg}_L L(z) = \operatorname{trdeg}_{K_{\exp}} K_{\exp}(z) = |D|$. Thus evaluation at z is injective on $L[Z]$. The equality $\det(P(z) - \rho I_N) = 0$ forces $\det(P(Z) - \rho I_N)$ to vanish identically. Substituting $Z = \mathbf{1}$ gives $(1-\rho)^N = 0$, so $\rho = 1$. In particular zero cannot be an eigenvalue.

Unitarity makes every $M_H(t)$ nonnegative and doubly stochastic. Moreover $M_H(-t) = M_H(t)^T$, so symmetric sampling makes the average real symmetric even when H has nonreal entries. For a support edge, $U_H(t)_{uv} = itH_{uv} + O(t^2)$, whence the averaged entry is positive; all diagonal entries are positive as well. Each support component consequently gives an irreducible nonnegative stochastic block with positive diagonal. Perron–Frobenius proves the remaining assertions. □

Corollary 6.2 (Entries, principal minors, and uniform mixing)**.** *Under Theorem 6.1, put* $A = \widehat{M}_{a,\nu}(H)$.

1. *If* A_{uv} *is algebraic over* $K_{\exp}$*, then* $A_{uv} = \delta_{uv}$*. Every entry strictly between* 0 *and* 1 *is therefore transcendental over* $K_{\exp}$.
2. *Every principal submatrix of* A *is nonsingular. Its determinant is transcendental over* $K_{\exp}$ *if its index set contains a nonisolated support vertex; otherwise that submatrix is the identity.*
3. *If* $N > 1$*, the average cannot be the uniform matrix* $N^{-1}\mathbf{1}\mathbf{1}^T$.

Proof. More generally, if a polynomial F in matrix entries has coefficients in $K_{\exp}$ and $F(A)$ is algebraic over $K_{\exp}$, the same injectivity argument applied to $F(P(Z)) - F(A)$ gives

$$F(A) = F(I_N). \tag{7}$$

This proves the assertion about entries. For a principal index set S, $\det P_S(\mathbf{1}) = 1$, so $\det P_S(z) \neq 0$. To decide whether this determinant polynomial is constant, let the half-width b tend to zero while H, ν remain fixed. Set $Q_{uv} = |H_{uv}|^2$ for $u \neq v$ and $Q_{uu} = -\sum_{v \neq u} |H_{uv}|^2$. Symmetry of the law and its variance $b^2/(2\nu + 2)$ yield

$$\widehat{M}_{b,\nu}(H) = I_N + \frac{b^2}{2\nu + 2} Q + O(b^4),$$
$$\det(\widehat{M}_{b,\nu}(H)_S) = 1 - \frac{b^2}{2\nu + 2} \sum_{u \in S} \sum_{v \neq u} |H_{uv}|^2 + O(b^4).$$

The coefficient is strictly negative exactly when S contains a nonisolated vertex. In that case $\det P_S(Z)$ is nonconstant, and a nonconstant polynomial in an independent tuple is transcendental over the coefficient field. If every vertex in S is isolated, the principal submatrix is the identity. Finally, the uniform matrix has rank one, whereas A has rank N. □

Remark 6.3 (The algebraic-width hypothesis). For the two-vertex path the average is

$$\frac{1}{2} \begin{pmatrix} 1 + \phi_{a,\nu}(2) & 1 - \phi_{a,\nu}(2) \\ 1 - \phi_{a,\nu}(2) & 1 + \phi_{a,\nu}(2) \end{pmatrix},$$

with eigenvalues 1 and $\phi_{a,\nu}(2)$. At a positive zero of $J_\nu(2a)$ it has rank one and is uniform; all such half-widths are transcendental by the independence input. For the arcsine law the second eigenvalue can also be negative, so the nonsingularity theorem does not follow from positive definiteness.

Remark 6.4 (Scope of the application). Coutinho–Guo [2, Open Problem 4.6] propose developing the spectral theory of random-time average mixing matrices. Theorem 6.1 settles rank and algebraic eigenvalue possibilities for the family (4) at algebraic widths; it does not settle their problem for arbitrary distributions or questions about the ordinary long-time average. The characteristic-function framework is due to Baptista–Coutinho–Marques [3]. For deterministic algebraic times, related exclusions over $\overline{\mathbb{Q}}$ already follow from Lindemann–Weierstrass. The present application uses continuous sampling and proves the stronger exclusion over $K_{\exp}$: deterministic finite-system mixing matrices have entries in $K_{\exp}$ and hence eigenvalues algebraic over that field. No assertion of joint independence of the averaged eigenvalues, or of historical priority for this application, is intended.

7 Arithmetic restrictions on Robin spectra of Euclidean balls

Let $k = \overline{\mathbb{Q}}$, let $\mathbf{E}$ be the ring of values of E-functions at algebraic arguments, and let $\overline{\mathbf{E}}$ be its integral closure in $\mathbb{C}$. Write $K_{\exp} = k(e^a : a \in k)$. We quote two further inputs from the unified manuscript [1, Theorems 22.2 and 34.2]:

$$\overline{\mathbf{E}} \cap k(C(t)) = k \quad (t \in k^\times), \qquad \overline{\mathbf{E}}^\times \cap K_{\exp} = \{ae^b : a \in k^\times,\ b \in k\}. \tag{8}$$

Here $C(t) = I_1(t)/I_0(t)$. The first equality is a subfield consequence of the complex one-block intersection theorem; the second follows from the complete Bessel-field unit theorem. Algebraic independence of $J_0(x), J_1(x)$ at nonzero algebraic x is also available, by putting $t = ix$ in the joint theorem.

Consider

$$-\Delta u = \lambda u \quad \text{in } B_R \subset \mathbb{R}^d, \qquad \partial_n u + \sigma u = 0 \quad \text{on } \partial B_R, \tag{9}$$

where $d \geq 2$ and $R > 0$ is algebraic. In the statements about the real spectrum, σ is real. The zero eigenvalue is explicitly excluded from transcendence assertions: it occurs when $R\sigma = -\ell$ for a nonnegative integer angular degree ℓ.

7.1 An exact even-dimensional obstruction

Lemma 7.1. *For $x \in k^\times$ and $n \in \mathbb{Z}_{\geq 0}$, $J_n(x) \neq 0$ and*

$$L_n(x) := \frac{xJ_n'(x)}{J_n(x)} \notin \overline{\mathbf{E}}.$$

In fact $k(L_n(x)) = k(C(ix))$ and $\overline{\mathbf{E}} \cap k(L_n(x)) = k$.

Proof. The recurrence

$$\begin{pmatrix} J_{n+1}(x) \\ J_{n+2}(x) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2(n+1)/x \end{pmatrix} \begin{pmatrix} J_n(x) \\ J_{n+1}(x) \end{pmatrix}$$

has determinant one. It expresses (J_n, J_{n+1}) as an invertible k-linear transformation of (J_0, J_1) at the fixed algebraic point. Independence of $J_0(x), J_1(x)$ gives nonvanishing of each $J_n(x)$. Moreover

$$L_n(x) = n - x\frac{J_{n+1}(x)}{J_n(x)}$$

is a nonconstant fractional-linear transformation of $J_1(x)/J_0(x) = C(ix)/i$. Thus the fields agree, and (8) gives the result. □

Theorem 7.2 (Even-dimensional Robin spectra)**.** *If d is even and $\sigma \in \overline{\mathbf{E}}$, every nonzero eigenvalue of (9) is transcendental.*

Proof. Suppose $\lambda \neq 0$ is algebraic and set $x = R\sqrt{\lambda} \in k^\times$. Separation into spherical harmonics gives, for some $\ell \geq 0$, a regular radial solution proportional to $r^{-(d-2)/2}J_n(xr/R)$, with $n = \ell + (d-2)/2 \in \mathbb{Z}_{\geq 0}$. The boundary equation is

$$\frac{xJ_n'(x)}{J_n(x)} = \frac{d-2}{2} - R\sigma.$$

Its right side lies in $\overline{\mathbf{E}}$, contradicting Lemma 7.1. The argument allows complex σ and complex λ as well. □

For a fixed nonzero algebraic spectral parameter, the angular eigenvalues of the Dirichlet-to-Neumann map on an even-dimensional ball are

$$\Lambda_\ell(\lambda) = \frac{1}{R}\left(\frac{xJ'_{\ell+(d-2)/2}(x)}{J_{\ell+(d-2)/2}(x)} - \frac{d-2}{2}\right).$$

Each generates the same field $k(C(ix))$. Consequently the field generated by any nonempty collection of these angular eigenvalues has intersection exactly k with $\overline{\mathbf{E}}$. This is an arithmetic assertion about the boundary response operator, stronger than the individual transcendence of its eigenvalues.

7.2 Positive eigenvalues in every dimension

Lemma 7.3 (Two-exponential logarithmic derivatives)**.** *Let $x \in k^\times$, and suppose a value and a derivative have the form*

$$f = ae^x + be^{-x}, \qquad f_1 = Ae^x + Be^{-x}, \qquad a, b, A, B \in k, \quad aB - bA \neq 0.$$

If $f \neq 0$ and $f_1/f \in \overline{\mathbf{E}}$, then $ab = 0$.

Proof. Put $r = f_1/f$. Inverting the coefficient matrix gives

$$\frac{e^x}{f} = \frac{B - br}{aB - bA} \in \overline{\mathbf{E}}.$$

Since $e^{-x} \in \mathbf{E}$, $1/f \in \overline{\mathbf{E}}$. Also $f \in \mathbf{E}$, so it is a unit of $\overline{\mathbf{E}}$ lying in $K_{\exp}$. By (8), $f = \alpha e^\beta$ with $\alpha \in k^\times$, $\beta \in k$. Lindemann–Weierstrass, applied to the distinct exponents among $x, -x, \beta$, then forces $ab = 0$. □

Theorem 7.4 (All positive Robin eigenvalues)**.** *For every $d \geq 2$ and every real $\sigma \in \overline{\mathbf{E}}$, all positive eigenvalues of (9) are transcendental.*

Proof. Only odd $d = 2q + 3$ remains. For angular degree ℓ, put $N = \ell + q$. The regular radial solution for a positive eigenvalue is $r^{-q}j_N(xr/R)$, where $x = R\sqrt{\lambda}$ and j_N is the spherical Bessel function. Thus

$$R\sigma = q - \frac{xj'_N(x)}{j_N(x)}.$$

At real algebraic $x > 0$, the finite Hankel expansions give $j_N(x) = ae^{ix} + be^{-ix}$, and its derivative has the same two exponents with algebraic coefficients. The determinant of these two coefficient rows is nonzero by the Hankel Wronskian. Moreover both a and b are nonzero: the outgoing and incoming Hankel functions are complex conjugates on the positive real axis, and a common zero would contradict their nonzero Wronskian. The boundary equation would put $j'_N(x)/j_N(x)$ in $\overline{\mathbf{E}}$, contradicting Lemma 7.3, with ix in place of x. □

7.3 A complete description of the odd-dimensional exceptions

For $N \geq 0$ define the explicit polynomials

$$a_{N,k} = \frac{(N+k)!}{2^k k!(N-k)!}, \qquad P_N(y) = \sum_{k=0}^{N} (-1)^k a_{N,k} y^{N-k}, \qquad Q_N(y) = \sum_{k=0}^{N} a_{N,k} y^{N-k}. \tag{10}$$

The modified spherical Bessel function is

$$i_N(y) = \frac{e^y P_N(y) + (-1)^{N+1} e^{-y} Q_N(y)}{2y^{N+1}}. \tag{11}$$

These are the terminating half-integer Hankel formulas; see DLMF §§10.47–10.49.

Theorem 7.5 (Exact negative algebraic exceptions)**.** *Let $d = 2q + 3$ and $\sigma \in \overline{\mathbf{E}} \cap \mathbb{R}$. An algebraic number $\lambda < 0$ is an eigenvalue of (9) if and only if, for some integer $N \geq q$, the number $y = R\sqrt{-\lambda} > 0$ satisfies*

$$P_N(y) = 0, \qquad R\sigma = q - y\frac{i_N'(y)}{i_N(y)}. \tag{12}$$

Every such N is odd, every such y is algebraic, and the corresponding Robin constant belongs to $\mathbf{E}$. More explicitly,

$$R\sigma = q + y + N + 1 - y\frac{Q_N'(y)}{Q_N(y)} - ye^{2y}\frac{P_N'(y)}{Q_N(y)}. \tag{13}$$

The coefficient of e^{2y} in this formula is nonzero, so the Robin constant of each exception is transcendental.

Proof. The regular radial solution is $r^{-q} i_N(yr/R)$ with $N = \ell + q$. The boundary equation is the second equality in (12). If λ is algebraic, then y is positive algebraic, and this equation puts $i_N'(y)/i_N(y)$ in $\overline{\mathbf{E}}$. The two exponential components in (11) are independent solutions of the modified spherical Bessel equation. Their Wronskian is nonzero at $y \neq 0$; hence the coefficient determinant for i_N, i_N' is nonzero. Lemma 7.3 now implies that one exponential coefficient vanishes. Since $Q_N(y) > 0$, it must be $P_N(y)$.

Conversely, a positive root of P_N is algebraic, and (11) is nonzero there because $Q_N(y) > 0$. The boundary constant defined in (12) therefore gives an eigenfunction of angular degree $N - q$. Positivity $i_N(y) > 0$, which follows from the series for $I_{N+1/2}$, implies that N is odd when $P_N(y) = 0$. Differentiating (11) at such a root gives (13). Its coefficients are algebraic, so $\sigma \in k[e^{2y}] \subset \mathbf{E}$.

Finally $P_N'(y) \neq 0$. Otherwise the component $e^y P_N(y)/(2y^{N+1})$ and its derivative would both vanish at the ordinary point $y > 0$ of its second-order differential equation. Uniqueness would make that component identically zero, a contradiction. Lindemann–Weierstrass consequently makes the displayed boundary constant transcendental. □

Example 7.6 (A necessary exception, not a numerical artifact). In the unit ball in $\mathbb{R}^3$, take $N = 1$. Then $P_1(y) = y - 1$, $Q_1(y) = y + 1$, and

$$\boxed{\sigma = \frac{5 - e^2}{2} \in \mathbf{E}, \qquad \lambda = -1.}$$

Indeed $i_1(1) = e^{-1}$ and $i_1'(1) = (e - 5e^{-1})/2$. The angular-degree-one eigenfunctions have the usual three-dimensional spherical-harmonic multiplicity. Thus the positive-spectrum theorem cannot be extended to all nonzero eigenvalues in odd dimensions.

7.4 Algebraic Robin parameters and accidental degeneracy

The next statement uses classical transcendence and a recurrence calculation; unlike Theorems 7.2–7.5, its proof does not need the new integral-intersection or unit results.

Theorem 7.7 (An algebraic-parameter Bourget statement)**.** *Let $\nu \in \{0, \frac{1}{2}\}$, $h \in k$, and let n, m be distinct nonnegative integers. The functions*

$$D_n(z) = zJ'_{\nu+n}(z) + hJ_{\nu+n}(z), \qquad D_m(z) = zJ'_{\nu+m}(z) + hJ_{\nu+m}(z)$$

have no common nonzero complex zero.

Proof. Write $J_{\nu+n} = A_n J_\nu + B_n J_{\nu+1}$ with $A_n, B_n \in \mathbb{Q}(z)$, using the usual recurrence. In particular these rational functions are Laurent polynomials with their only possible finite pole at zero. The transfer matrices have determinant one. Write $D_n = a_n J_\nu + b_n J_{\nu+1}$. We first check that

$$\Delta_{n,m}(z) = a_n(z)b_m(z) - b_n(z)a_m(z)$$

is not identically zero for any h.

Induction in the recurrence gives the following Laurent expansions at infinity. If n is even and $\epsilon_n = (-1)^{n/2}$, then

$$(a_n, b_n) = \epsilon_n \left(h + \nu - \nu n - \frac{n^2}{2} + O(z^{-2}), \quad -z + O(z^{-1}) \right).$$

If n is odd and $\epsilon_n = (-1)^{(n-1)/2}$, then

$$(a_n, b_n) = \epsilon_n \left(z + O(z^{-1}), \quad h - \nu n - \frac{n^2+1}{2} + O(z^{-2}) \right).$$

For clarity, the induction follows from

$$A_{n+1} = \frac{2(\nu+n)}{z} A_n - A_{n-1}, \qquad B_{n+1} = \frac{2(\nu+n)}{z} B_n - B_{n-1},$$

starting with $(A_0, B_0) = (1, 0)$ and $(A_1, B_1) = (0, 1)$, and then using $(a_n, b_n) = (\nu+n+h)(A_n, B_n) - z(A_{n+1}, B_{n+1})$. Consequently, for n, m of the same parity,

$$\Delta_{n,m}(z) = \epsilon_n \epsilon_m \frac{(\nu+n)^2 - (\nu+m)^2}{2} z + O(z^{-1}),$$

whose leading coefficient is nonzero. For opposite parities its leading term is $\pm\epsilon_n\epsilon_m z^2$. Thus $\Delta_{n,m} \neq 0$, independently of h.

If $z_0 \neq 0$ were a common zero, the nonzero vector $(J_\nu(z_0), J_{\nu+1}(z_0))$ would lie in the kernel of the coefficient matrix. The vector cannot be zero, by the recurrence and uniqueness for Bessel's differential equation. Therefore $\Delta_{n,m}(z_0) = 0$, so z_0 is algebraic when h is algebraic.

For $\nu = 0$, independence of $J_0(z_0), J_1(z_0)$ contradicts $D_n(z_0) = 0$: the row $(a_n(z_0), b_n(z_0))$ is nonzero since the transfer matrix is invertible and $z_0 \neq 0$. For $\nu = \frac{1}{2}$, use the elementary sine and cosine expressions for $J_{1/2}, J_{3/2}$. Their coefficient matrix in e^{iz_0}, e^{-iz_0} is invertible, up to a common nonzero factor. Thus $D_n(z_0) = 0$ would be a nontrivial algebraic linear relation between e^{iz_0} and e^{-iz_0}, contradicting Lindemann–Weierstrass. □

Corollary 7.8 (Removal of genericity for algebraic Robin constants)**.** *For a Euclidean ball of algebraic radius and every real algebraic Robin constant, there are no accidental coincidences between distinct spherical-harmonic degrees. Hence every eigenfunction is a separated radial factor times a spherical harmonic, and the Robin Pleijel constant equals the Dirichlet Pleijel constant by the theorem of Hassannezhad–Sher. In dimension two this common constant is approximately* 0.461.

Proof. The radial boundary condition has the form in Theorem 7.7 with $h = R\sigma - (d-2)/2$ and with integer or half-integer orders. The common-zero theorem excludes all nonzero coincidences. At zero energy the boundary condition on a harmonic homogeneous polynomial of degree ℓ is $R\sigma = -\ell$, so distinct degrees cannot coincide there either. Apply Hassannezhad–Sher, Theorem 1.4. □

7.5 Known status and the actual remaining obstruction

Hassannezhad–Sher [5], Assumption 1.3 and Theorem 1.4, prove the ball Pleijel equality under absence of accidental coincidences. Their Proposition 3.2 makes that condition generic, and Remark 3.3 gives a real Robin parameter for which it fails. Their Open Question 1.5 asks for boundary-condition independence on general domains. Corollary 7.8 supplies an explicit arithmetic subclass; it does not settle that general-domain question. Wang [6], Remark 1.3, explicitly records the disk noncoincidence result for algebraic Robin constants less than one. The calculation above has no such restriction. A literature search has not established priority for this unrestricted elementary extension, so it should not be presented as the solution of a newly discovered named conjecture.

The clamped round-plate problem is different. Its characteristic functions are $W_n(x) = I_n'(x)J_n(x) - I_n(x)J_n'(x)$, and the desired multiplicity-two bound would rule out a common positive zero of two distinct W_n. Mangoubi–Rosenblatt [7], Conjecture 1.3, leaves precisely this problem open and proves multiplicity at most four in dimension two. A common zero is already transcendental. The unified paper's arithmetic value theorems apply at algebraic arguments and therefore cannot simply be evaluated at that zero. Their second-order recurrence produces a polynomial relation between the common zero and a Bessel quotient at that zero, not a polynomial over k in the zero alone. No removal of this obstruction is proved here.

Likewise, for an $\overline{\mathbf{E}}$-valued Robin parameter, the recurrence determinant makes a hypothetical common wavenumber integral over $\overline{\mathbf{E}}$ after clearing a power of z: its leading coefficient is a nonzero algebraic constant. It need not make that wavenumber algebraic. Applying an algebraic-argument value theorem at such a number would be invalid. Thus the no-accidental-degeneracy statement above is asserted for algebraic parameters, not for every parameter in $\overline{\mathbf{E}}$.

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